

MOHNISH KODUKULLA

Software Engineer | AI Systems | Google Summer of Code 2026 @ Debezium
Chennai, India | kmohnishm@gmail.com | LinkedIn | Portfolio | GitHub

EDUCATION

B.Tech in Computer Science and Engineering

Aug 2023 – Present

Vellore Institute of Technology, Chennai

GPA: 9.28/10

Relevant Coursework: Data Structures & Algorithms, Design and Analysis of Algorithms, Operating Systems, Computer Networks, Database Systems, Software Engineering, Artificial Intelligence, Cloud Architecture

SKILLS

Languages: Python, JavaScript/TypeScript, C++, Java, SQL, Shell Scripting

Backend: FastAPI, Django, Flask, Node.js, PostgreSQL, Redis, Celery, SQLAlchemy, WebSockets, Supabase

AI/ML: LangChain, LangGraph, PyTorch, Scikit-learn, RAG, ChromaDB, PGVector, Milvus, Agentic AI

Cloud & DevOps: Docker, AWS EC2, Prometheus, Grafana, Alertmanager, Git, GitHub Actions

Frontend: React.js, Next.js, Vite, Tailwind CSS

WORK EXPERIENCE

Debezium

May 2026 - Present

GSoC Contributor - AI Infrastructure & Developer Tooling

Remote

- Designed and built **PyDebeziumAI**, an open-source Python library that streams relational database changes into LangChain and LangGraph applications through pluggable vector-store adapters
- Implemented change-data-capture pipelines that automatically synchronize vector embeddings with relational database updates, ensuring RAG applications remain consistent with source data .

SRIP - VIT Chennai

May 2025 - Jul 2025

Summer Research Intern - AI Systems & Full-Stack Development

Chennai, India

- Developed two AI-assisted learning platforms used during a student pilot, combining retrieval-augmented generation, automated evaluation, and adaptive quiz generation across 150+ sessions
- Built a multi-stage inference pipeline for code evaluation, contextual hint generation, and automated scoring while reducing redundant LLM requests by 35% through response caching

PROJECTS

AWS Cloud - Healthcare Network Monitor

[View Project](#)

- Built a cloud-native monitoring platform of five containerized services on AWS EC2 with Prometheus-based telemetry, Grafana dashboards, and Alertmanager threshold notifications
- Trained an anomaly detection model on 3,000+ telemetry samples, achieving 92% precision for abnormal vital-sign detection with <2s alert latency

Vyaapar.AI - Blockchain Investment Platform

[View Project](#)

- Developed a full-stack blockchain investment platform featuring wallet authentication, NFT-based company verification, smart-contract investment tracking, and real-time portfolio analytics
- Built on React, Node.js/Express, Supabase, Ethers.js, and Socket.io; on-chain transactions via Solidity contracts; **Runner-Up, ILH Hackathon**

Coding Platform - Competitive Programming

[View Project](#)

- Built a competitive programming platform with browser-based code execution (Monaco Editor + Piston), retrieval-augmented AI hints via LangChain, user authentication, analytics dashboards, and PostgreSQL-backed submission tracking for 500+ community attempts

SALS - Smart Adaptive Learning System

[View Project](#)

- Adaptive learning platform with LLM-powered quiz generation; improved weak-topic detection accuracy by 30% across 500+ attempts; Django REST backend with React frontend deployed on Vercel

OPEN SOURCE

Debezium - GSoC 2026

[debezium-ai-python](#) · [debezium-examples](#) · [debezium-quarkus](#) · [debezium](#)

- Contributed production code across four repositories spanning AI infrastructure, CDC integrations, documentation, connector improvements, and Quarkus tooling

OpenROAD (VLSI Physical Design)

[Pull Requests](#)

- Merged **dbSta** clock signal propagation fix and **odb** 3D block-vector coordinate validation into the production VLSI toolchain

ACHIEVEMENTS

- Selected as a **Google Summer of Code 2026** contributor with Debezium to develop AI infrastructure for real-time change-data-capture pipelines
- Runner-Up, ILH Hackathon** - Vyaapar.AI, a blockchain-based startup investment and portfolio platform
- Development Lead, CYSCOM VIT Chennai:** led and mentored engineering team, shipped production tools serving 300+ members,
- Solved 300+ problems on LeetCode spanning dynamic programming, graphs, trees, and binary search
- IBM Skills Network (Cognitive Class)** - Successfully completed *IBM GEN AI ADV*